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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Build a SOHO network

Task:

❖ **1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).**

➢ **Aim**

- To design a Small Office/Home Office (SOHO) network using one broadband Internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer, showing the type of connection used by each device.

➢ **Introduction**

- A SOHO (Small Office/Home Office) network is a small computer network designed for use in a home or a small office. It allows multiple devices to share an Internet connection and communicate with each other.
- A SOHO network generally uses a broadband Internet connection and a Wi-Fi router to provide both wired and wireless connectivity.

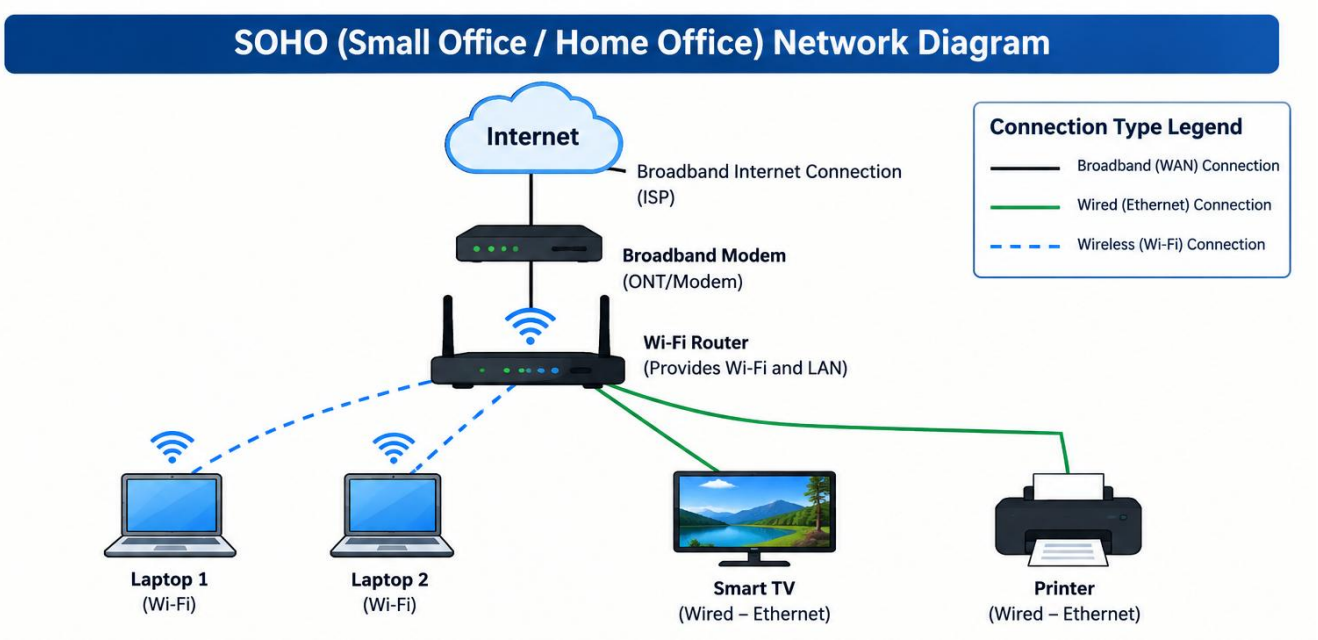
- In this network, two laptops connect wirelessly through Wi-Fi, while the smart TV and printer connect to the router using Ethernet cables.

➢ **Requirements**

➢ The following devices are required to build this SOHO network:

1. One broadband Internet connection
2. One broadband modem/ONT
3. One Wi-Fi router
4. Two laptops
5. One smart TV
6. One network printer
7. Ethernet cables for wired connections

➢ **SOHO Network Diagram**



- **Explanation of Devices and Connections**

Device	Connection Type	Description
Broadband Internet	Wired (WAN)	Provides Internet connectivity through the Internet Service Provider (ISP).
Wi-Fi Router	Wired to modem	Receives the Internet connection and distributes it to network devices.
Laptop 1	Wireless (Wi-Fi)	Connects to the router wirelessly.
Laptop 2	Wireless (Wi-Fi)	Connects to the router wirelessly.
Smart TV	Wired (Ethernet)	Connects to the router using an Ethernet cable.
Printer	Wired (Ethernet)	Connects to the router using an Ethernet cable.

- **Working of the SOHO Network**
  1. The broadband Internet connection is provided by the Internet Service Provider (ISP).
  2. The broadband connection reaches the modem or ONT.
  3. The modem/ONT connects to the Wi-Fi router through an Ethernet cable.

4. The Wi-Fi router distributes the Internet connection to all connected devices.
5. Both laptops access the network wirelessly using Wi-Fi.
6. The smart TV and printer connect to the router through wired Ethernet connections.
7. All connected devices can access the Internet through the same broadband connection.

- **Advantages of a SOHO Network**
  1. **Internet Sharing:** Multiple devices can use one broadband Internet connection.
  2. **Wireless Connectivity:** Laptops can connect without using network cables.
  3. **File and Resource Sharing**: Connected devices can share available network resources.
  4. **Printer Sharing:** Multiple devices can use the same network printer.
  5. **Easy Management:** The Wi-Fi router makes it easier to manage network connections.
  6. **Cost Effective:** It provides networking facilities without requiring a large office network.

- **Result**
- A SOHO network was successfully designed using one broadband Internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. The laptops were connected wirelessly, while the smart TV and printer were connected through wired Ethernet connections.

❖ **2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.**

➢ **Aim**

- To identify the hardware components, cables, and networking devices required to physically build the designed SOHO (Small Office/Home Office) network, along with suitable brand and model suggestions.

➢ **Introduction**

- A SOHO network requires different hardware components to connect computers, entertainment devices, and printers to the Internet. These components include a broadband modem or ONT, Wi-Fi router, laptops, smart TV, printer, and Ethernet cables.
- For this setup, two laptops connect wirelessly through Wi-Fi, while the smart TV and printer connect to the router using wired Ethernet connections.

➢ List of Required Hardware Components

Sr. No.	Hardware Component	Suggested Brand/Model	Quantity	Purpose
1	Broadband Modem/ONT	ISP-provided Nokia or Huawei ONT	1	Receives the broadband Internet connection from the ISP.
2	Wi-Fi Router	TP-Link Archer C6 AC1200	1	Provides wireless and wired network connectivity.
3	Laptop	HP, Dell, or Lenovo laptop	2	Connects to the network using Wi-Fi.
4	Smart TV	Samsung or LG Smart TV	1	Connects to the network using an Ethernet cable.
5	Network Printer	HP or Brother printer with an Ethernet port	1	Allows computers to print through the network.
6	Cat6 Ethernet Cable	D-Link, TP-Link, or other	2	Connects the modem to the

		reliable Cat6 cable		router and the router to wired devices.
7	RJ-45 Connectors	D-Link or compatible RJ-45 connectors	As required	Used for terminating Ethernet cables when making custom cables.
8	Power Adapters	Supplied with networking devices	As required	Provide electrical power to the modem and router.

- **Suggested Brands and Models**
- **1. Broadband Modem/ONT**
  - Suggested brand: Nokia or Huawei.
  - Quantity: 1.
  - The ONT/modem is generally provided by the Internet Service Provider (ISP).
  - It receives the broadband connection and provides an Ethernet connection to the Wi-Fi router.
- **2. Wi-Fi Router**
  - Suggested model: TP-Link Archer C6 AC1200.
  - Quantity: 1.
  - It supports dual-band Wi-Fi (2.4 GHz and 5 GHz).

- It provides one Gigabit WAN port and four Gigabit LAN ports.
- It can connect wireless devices and wired devices simultaneously.

➢ **3. Laptops**

- Suggested brands: HP, Dell, or Lenovo.
- Quantity: 2.
- Both laptops connect to the Wi-Fi router wirelessly.
- They can access the Internet and shared network resources.

➢ **4. Smart TV**

- Suggested brands: Samsung or LG.
- Quantity: 1.
- The smart TV connects to the router using a Cat6 Ethernet cable.
- A wired connection provides stable network connectivity for online streaming.

➢ **5. Network Printer**

- Suggested brands: HP or Brother.
- Quantity: 1.
- A printer with an Ethernet (RJ-45) network port should be selected.
- It connects to the Wi-Fi router through an Ethernet cable and can be shared by both laptops.

➢ **Cables and Accessories Required**

➢ The following cables and accessories are required:

1) **Cat6 Ethernet Cable:** Used to connect the broadband modem/ONT to the router and the router to the smart TV and printer.
2) **RJ-45 Connectors:** Used at the ends of Ethernet cables when preparing custom cables.

3) **Power Adapters:** Required to provide power to the modem/ONT and Wi-Fi router.
4) **Power Extension Board:** Can be used if multiple devices need power sockets in one location.

- For this network, two Ethernet cable connections are required:
  - Modem/ONT to Wi-Fi router (WAN port).
  - Wi-Fi router to smart TV and printer (two separate LAN connections).

- **Need for a Network Switch or Wi-Fi Extender**

- **Network Switch**
- A separate network switch is not required for this SOHO setup because the suggested TP-Link Archer C6 router provides four Gigabit LAN ports. Two LAN ports are sufficient for connecting the smart TV and printer, while the remaining ports are available for future devices.
- If more wired devices are added later, an additional switch such as the **TP-Link TL-SG1008D 8-Port Gigabit Switch** can be used.

- **Wi-Fi Extender**
- A Wi-Fi extender is not required if the router provides sufficient wireless coverage for both laptops. However, if the home or office has weak Wi-Fi signal areas, a compatible range extender such as a **TP-Link RE305** can be considered.

- **Estimated Hardware Connection Arrangement**

Connection	Connection Type
ISP to Modem/ONT	Broadband connection
Modem/ONT to Wi-Fi Router	Wired (Ethernet/WAN)
Laptop 1 to Wi-Fi Router	Wireless (Wi-Fi)
Laptop 2 to Wi-Fi Router	Wireless (Wi-Fi)
Smart TV to Wi-Fi Router	Wired (Ethernet/LAN)
Printer to Wi-Fi Router	Wired (Ethernet/LAN)

- **Online Product Listing References**
- The following online listings and manufacturer information can be used as references for the suggested hardware:
  - TP-Link Archer C6 AC1200 Router – Flipkart listing.
  - TP-Link Archer C6 – Official manufacturer specifications.
  - TP-Link TL-SG1008D – 8-Port Gigabit Switch model reference.
- Product prices and availability may change depending on the seller, location, and time of purchase. The exact ONT model also depends on the Internet Service Provider.

- **Result**
- The hardware components required to physically build the SOHO network were identified. Suitable brand and model suggestions were selected for the

modem/ONT, Wi-Fi router, laptops, smart TV, printer, and Ethernet cables. The need for an additional network switch or Wi-Fi extender was also evaluated.

❖ **3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.**

➢ **Aim**

- To configure a Wi-Fi router for a SOHO (Small Office/Home Office) network by setting a unique SSID, enabling WPA2 security, and changing the default administrator password to improve network security.

➢ **Introduction**

- A Wi-Fi router is an important networking device that connects multiple devices to the Internet and allows wireless communication.
- For a SOHO network, the router must be configured properly to provide secure and reliable connectivity. Important configuration steps include setting a unique **SSID (Service Set Identifier)**, enabling **WPA2 wireless security**, and changing the default administrator password.
- In this practical, a TP-Link Archer C6 Wi-Fi router is considered as an example.

- **Requirements**
- The following are required for router configuration:
  1. One Wi-Fi router (TP-Link Archer C6 or similar).
  2. A laptop or computer.
  3. An Ethernet cable or an existing Wi-Fi connection.
  4. A web browser.
  5. Internet connection.
  6. Router login credentials.

- **Step-by-Step Instructions to Configure the Wi-Fi Router**
- **Step 1: Connect to the Wi-Fi Router**
  1. Connect the laptop to the router using an Ethernet cable or its existing Wi-Fi network.
  2. Switch on the Wi-Fi router.
  3. Make sure the laptop is connected to the router successfully.
- **Step 2: Open the Router Administration Page**
  1. Open a web browser on the laptop.
  2. Enter the router's administration address in the address bar.
  3. For many TP-Link routers, the address is 192.168.0.1 or tplinkwifi.net.
  4. Press Enter to open the router login page.
  5. Enter the current administrator login credentials.
- **Step 3: Set a Unique SSID**
  1. After logging in, open the **Wireless Settings** section.
  2. Find the **Wireless Network Name (SSID)** option.
  3. Enter a unique network name.
- **Example SSID:** Prem_SOHO_Network
  4. Save the changes.

- A unique SSID helps users identify their own wireless network easily.
- **Step 4: Enable WPA2 Security**
  1. Open the **Wireless Security** section.
  2. Find the security mode option.
  3. Select **WPA2-Personal (WPA2-PSK)**.
  4. Select **AES** as the encryption method, if the option is available.
  5. Enter a strong Wi-Fi password.
- **Example password format:** A long combination of uppercase and lowercase letters, numbers, and special characters.
- Save the wireless security settings.
- WPA2 helps protect the wireless network by encrypting communication and preventing unauthorized users from easily accessing the network.
- **Step 5: Change the Default Administrator Password**
  1. Open the **System Tools**, **Administration**, or **Change Password** section.
  2. Find the administrator password settings.
  3. Enter the current administrator password.
  4. Enter a new, strong administrator password.
  5. Confirm the new password.
  6. Click Save or Apply.
- Changing the default administrator password prevents unauthorized users from easily accessing the router's configuration settings.
- **Step 6: Save and Apply the Configuration**
  1. Review the SSID and WPA2 security settings.
  2. Click **Save** or **Apply**.
  3. The router may restart or disconnect wireless devices temporarily.
  4. Reconnect the laptop using the new Wi-Fi network name and password.
- **Step 7: Verify the Configuration**

1. Search for the new SSID on the laptop.
2. Select Prem_SOHO_Network.
3. Enter the newly configured Wi-Fi password.
4. Confirm that the laptop connects successfully.
5. Open the router administration page again and verify that the settings have been saved.
6. Check that the new administrator password is accepted when logging in.

- **Example Router Configuration**

Configuration	Example Setting
Router Model	TP-Link Archer C6
SSID	Prem_SOHO_Network
Security Mode	WPA2-Personal (WPA2-PSK)
Encryption	AES
Wi-Fi Password	A strong, unique password
Administrator Password	A new, strong password

- **Security Recommendations**
  1. Always use a strong and unique Wi-Fi password.
  2. Change the default router administrator password.
  3. Use WPA2-AES security or WPA3 if supported by all required devices.
  4. Keep the router firmware updated.
  5. Disable WPS if it is not required.

6. Do not share the Wi-Fi password with unauthorized users.
7. Regularly check the list of connected devices.

➢ **Result**

➢ The Wi-Fi router was configured for the SOHO network by setting a unique SSID, enabling WPA2-Personal security with AES encryption, and changing the default administrator password. The wireless connection was then verified using a laptop.

❖ **4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**

➢ **Aim**

- To configure IP addresses for all devices in a SOHO (Small Office/Home Office) network by assigning static IP addresses to the printer and smart TV, while using DHCP to automatically assign IP addresses to both laptops.

➢ **Introduction**

- IP addressing is an important part of computer networking. Every device connected to a network requires an IP address to communicate with other devices.
- There are two common methods of assigning IP addresses:

1. **Static IP Address:** An IP address manually configured on a device. It generally remains the same until it is changed manually.
2. **Dynamic IP Address (DHCP):** An IP address automatically assigned to a device by a DHCP server, usually running on the Wi-Fi router.

- In this SOHO network, the printer and smart TV use static IP addresses so they can be accessed easily using consistent addresses. Both laptops use DHCP to receive their IP addresses automatically.

- **Network Configuration Details**
- The following network configuration is used:

Configuration	Value
Network Address	192.168.1.0
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DHCP Server	Wi-Fi Router
DHCP Address Pool	192.168.1.100 – 192.168.1.199
DNS Server	8.8.8.8

- **IP Address Assignment Table**
- The following table shows the MAC addresses and IP addresses assigned to all devices in the SOHO network.

Sr. No.	Device	MAC Address (Example)	IP Address	Assignment Method
1	Wi-Fi Router	02:10:20:30:40:01	192.168.1.1	Static
2	Laptop 1	02:10:20:30:40:02	192.168.1.100	DHCP
3	Laptop 2	02:10:20:30:40:03	192.168.1.101	DHCP
4	Smart TV	02:10:20:30:40:04	192.168.1.10	Static
5	Network Printer	02:10:20:30:40:05	192.168.1.20	Static

- **Step-by-Step IP Configuration**
- **Step 1: Configure the Wi-Fi Router**

1. Connect a laptop to the Wi-Fi router.
2. Open a web browser.
3. Enter the router's administration address, such as 192.168.1.1.
4. Log in using the administrator credentials.
5. Open the LAN or Network Settings section.
6. Set the router's LAN IP address to 192.168.1.1.
7. Open the DHCP Server settings.
8. Enable DHCP.

9. Set the DHCP address pool from 192.168.1.100 to 192.168.1.199.
10. Save the configuration.

- **Step 2: Configure Static IP for the Smart TV**
  1. Open the Network Settings on the Smart TV.
  2. Select Wired Network or Ethernet.
  3. Open IP Settings.
  4. Change the IP configuration from Automatic (DHCP) to Manual.
  5. Enter the following details:

Setting	Value
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

- Save the settings and reconnect the network.

- **Step 3: Configure Static IP for the Printer**
  1. Open the printer's Network Settings.
  2. Select the Ethernet or Wired Network option.
  3. Open IPv4 or TCP/IP Settings.
  4. Change the IP assignment method from Automatic (DHCP) to Manual or Static.
  5. Enter the following details:

Setting	Value
IP Address	192.168.1.20
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

- Save the configuration.
- Restart the printer if required.

- **Step 4: Configure DHCP for Laptop 1**
  1. Open the Network Settings on Laptop 1.
  2. Select the Wi-Fi connection.
  3. Open the IPv4 properties.
  4. Select **Obtain an IP address automatically**.
  5. Select **Obtain DNS server address automatically**.
  6. Save the settings.
  7. The router's DHCP server automatically assigns an available IP address, such as 192.168.1.100.

- **Step 5: Configure DHCP for Laptop 2**
  1. Open the Network Settings on Laptop 2.
  2. Select the Wi-Fi connection.
  3. Open the IPv4 properties.
  4. Select **Obtain an IP address automatically**.
  5. Select **Obtain DNS server address automatically**.

6. Save the settings.
7. The router's DHCP server automatically assigns an available IP address, such as 192.168.1.101.

- **Verification of IP Addresses**
- After configuring all devices, verify their IP addresses.
- **On Windows Laptops**
    a. Open Command Prompt.
    b. Type the following command:

ipconfig /all

    c. Press Enter.
    d. Check the IPv4 address, subnet mask, default gateway, and physical address (MAC address).

- **On the Smart TV and Printer**

1. Open the Network Settings.
2. Check the IP address and other network details.
3. Confirm that the configured static IP address is displayed.

- **Connectivity Test**
- Open Command Prompt on a laptop and use the following commands:
- **Test the router:**
- Ping 192.168.1.1
- **Test the smart TV:**
- Ping 192.168.1.10
- **Test the printer**
- Ping 192.168.1.20

- Successful ping replies indicate that the destination device is reachable on the network.

- **Important Points**
- Every device must have a unique IP address.
- Static IP addresses should be outside the DHCP address pool to avoid address conflicts.
- Both laptops receive IP addresses automatically through DHCP.
- The printer and smart TV use manually configured static IP addresses.
- All devices must use the correct subnet mask and default gateway.
- MAC addresses are unique hardware identifiers in real devices. The example MAC addresses in this assignment are illustrative only.

- **Result**
- IP addresses were assigned to all devices in the SOHO network. Static IP addresses were configured for the smart TV and network printer, while both laptops were configured to obtain IP addresses automatically through DHCP.

❖ **5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.**

➢ **Aim**

- To understand how to troubleshoot Internet connectivity problems when one laptop cannot access the Internet in a SOHO (Small Office/Home Office) network by identifying possible causes and applying suitable troubleshooting steps.

➢ **Introduction**

- A SOHO network allows multiple devices, such as laptops, smart TVs, and printers, to connect to the Internet through a common Wi-Fi router.
- Sometimes, one laptop may fail to connect to the Internet even when other devices are working properly. This problem can occur due to incorrect Wi-Fi settings, weak wireless signals, incorrect IP configuration, DNS problems, or router-related issues.
- Network troubleshooting is the process of identifying, diagnosing, and resolving network connectivity problems.

➢ **Possible Causes and Their Solutions**

➢ **Cause 1: Incorrect Wi-Fi Password or Connection Settings**

➢ **Problem:**

➢ The laptop may be unable to connect to the Wi-Fi network because an incorrect password has been entered or the wireless connection settings are incorrect.

- **Diagnosis and Solution:**
  1. Check whether Wi-Fi is enabled on the laptop.
  2. Verify that the correct Wi-Fi network name (SSID) is selected.
  3. Enter the correct Wi-Fi password.
  4. If necessary, forget the saved Wi-Fi network.
  5. Connect to the network again by entering the correct password.
  6. Check whether the laptop is successfully connected.
- **Expected Result:** The laptop should connect successfully to the Wi-Fi router.

- **Cause 2: Weak Wi-Fi Signal or Wireless Interference**
- **Problem:**
- The laptop may be located too far from the Wi-Fi router, or walls and other electronic devices may interfere with the wireless signal.
- **Diagnosis and Solution:**
  1. Check the Wi-Fi signal strength on the laptop.
  2. Move the laptop closer to the Wi-Fi router.
  3. Make sure the router is placed in an open and central location.
  4. Disconnect unnecessary wireless devices if the network is overloaded.
  5. Try connecting to another supported Wi-Fi band, such as 2.4 GHz or 5 GHz.
  6. Check whether the Internet connection works properly.
- **Expected Result:** A stronger Wi-Fi signal should improve the wireless connection.

- **Cause 3: Incorrect IP Address Configuration or DHCP Problem**
- **Problem:**

- The laptop may not receive a valid IP address from the router because DHCP is disabled or there is an IP configuration problem.
- **Diagnosis and Solution:**
  1. Open Command Prompt on the laptop.
  2. Enter the following command:
     ipconfig
  3. Check the IPv4 address, subnet mask, and default gateway.
  4. If the laptop has an incorrect or automatic private address, check the router's DHCP settings.
  5. Make sure DHCP is enabled on the Wi-Fi router.
  6. Enter the following commands to renew the IP address:
     ipconfig /release
     ipconfig /renew
  7. Check the IP configuration again using:
     ipconfig
- **Expected Result:** The laptop should receive a valid IP address from the router's DHCP server, such as 192.168.1.100.

- **Cause 4: DNS Server Problem**
- **Problem:**
- The laptop may be connected to the Wi-Fi network but unable to open websites because of an incorrect or unavailable DNS server.
- **Diagnosis and Solution:**
  Open Command Prompt.
  Ping the router using:
- **ping 192.168.1.1**
- If the router responds, test Internet connectivity using a public IP address:

- **ping 8.8.8.8**
- Check domain name resolution using:
- **nslookup google.com**
- If the IP address responds but domain names do not resolve, the problem may be related to DNS.
- Check the DNS settings and use a reliable DNS server, such as 8.8.8.8, if appropriate.
- Try opening the website again.
- **Expected Result:** DNS resolution should work correctly, allowing the laptop to access websites using domain names.

- **Cause 5: Router or Internet Service Provider (ISP) Problem**
- **Problem:**
- The Wi-Fi router may have a temporary issue, or the Internet Service Provider may be experiencing a network outage.
- **Diagnosis and Solution:**
  1. Check whether other devices, such as the second laptop or smart TV, can access the Internet.
  2. Check the router's Internet or WAN indicator.
  3. Restart the Wi-Fi router if required.
  4. Wait for the router to reconnect to the Internet.
  5. Test the Internet connection again.
  6. If all devices still cannot access the Internet, contact the Internet Service Provider.
- **Expected Result:** If the problem was caused by a temporary router issue, restarting the router may restore Internet connectivity. If the ISP connection is down, the service provider must resolve the issue.

- **Useful Network Troubleshooting Commands**
- The following Windows commands can help diagnose Internet connectivity problems:

Command	Purpose
ipconfig	Displays the current IP configuration.
ipconfig /release	Releases the current DHCP-assigned IP address.
ipconfig /renew	Requests a new IP address from the DHCP server.
ping 192.168.1.1	Tests connectivity to the Wi-Fi router.
ping 8.8.8.8	Tests connectivity to a public IP address.
nslookup google.com	Checks DNS name resolution.
tracert google.com	Displays the route taken by packets towards a destination.

- **Recommended Troubleshooting Sequence**
- The following sequence can be followed to identify the problem systematically:
- Check whether Wi-Fi is enabled and the correct network is selected.
- Check the wireless signal strength.
- Verify the laptop's IP address using ipconfig.
- Test connectivity to the router using ping 192.168.1.1.

- Test Internet connectivity using ping 8.8.8.8.
- Check DNS resolution using nslookup google.com.
- Restart the router if required or contact the ISP if the problem affects all devices.

- **Result**
- Different possible causes of Internet connectivity problems in a SOHO network were identified and suitable troubleshooting methods were studied. The use of Wi-Fi settings, IP configuration, DHCP, DNS, ping, and traceroute commands helps diagnose and resolve network problems.